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PATTERN TRANSFER PRINTING OF MULTI-LAYERED FEATURES

Abstract

Pattern transfer source substrates and printing methods are described, providing multi-layer paste stack elements like conductor lines or bumps, so that each stack element is printed on a receiving substrate in a single illumination step. The paste is filled layer-by-layer, possibly having different materials in different layers, with layer thickness controlled by parameters of the filling elements, e.g., in case of blades, the pressure, angle, velocity and flexibility (material) of the blade. Specifically, a bottom layer of the stack may be configured to interface the receiving substrate while one or more top layers may be configured to optimize the quality of the printed features. For example, bottom layers may be configured to bind to the substrate, to modify the substrate and/or provide a barrier from top layer(s) which may not be compatible with the substrate. In case of bumps, different layers may include adhesive and soldering materials.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation-In-Part of U.S. application Ser. No. 17/562,360, filed on Dec. 27, 2021, which is incorporated herein by reference in its entirety.

1. TECHNICAL FIELD

[0002] The present invention relates to the field of transfer printing, and more particularly, to printing multi-layered features.

2. DISCUSSION OF RELATED ART

[0003] U.S. Pat. No. 9,616,524, which is incorporated herein by reference in its entirety, teaches a method of depositing a material on a receiving substrate, the method comprising: providing a source substrate having a back surface and a front surface, the back surface carrying at least one piece of coating material; providing a receiving substrate positioned adjacent to the source substrate and facing the coating material; and radiating light towards the front surface of the source substrate, to remove at least one piece of the coating material from the source substrate and deposit said removed at least one piece onto the receiving substrate as a whole.

[0004] Lossen et al. (2015), Pattern Transfer Printing (PTP™) for c-Si solar cell metallization, 5^{sup}.th Workshop on Metallization for Crystalline Silicon Solar Cells, Energy Procedia 67:156-162, which is incorporated herein by reference in its entirety, teaches pattern transfer printing (PTP™) as a non-contact printing technology for advanced front side metallization of c-Si PV solar cells, which is based on laser-induced deposition from a polymer substrate.

SUMMARY OF THE INVENTION

[0005] The following is a simplified summary providing an initial understanding of the invention. The summary does not necessarily identify key elements nor be used to limit the scope of the invention, but merely serves as an introduction to the following description.

[0006] One aspect of the present invention provides a pattern transfer method comprising: filling, layer-by-layer, a plurality of stack elements that are each made of a plurality of printing paste layers into a plurality of dents on a source substrate, wherein the stack elements are filled into corresponding dents that are arranged in a pattern, and releasing the plurality of stack elements from the corresponding plurality of dents onto a receiving substrate simultaneously, in a single illumination step by a laser beam, forming transferred stack bumps on the receiving substrate, wherein at least two of the paste layers comprise different material compositions.

[0007] These, additional, and/or other aspects and/or advantages of the present invention are set forth in the detailed description which follows; inferable from the detailed description; and/or learnable by practice of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a better understanding of embodiments of the invention and to show how the same may be carried into effect, reference will now be made, purely by way of example, to the accompanying drawings in which like numerals designate corresponding elements or sections throughout.

[0009] In the accompanying drawings:

[0010] FIGS. 1, 2A and 2B are high-level schematic illustrations of pattern transfer sheets with multi-layered stacks of printing paste and pattern transfer methods, according to some embodiments of the invention.

[0011] FIG. 2C is a high-level schematic illustration of a combination of paste filling heads including doctor blade dispensing and recirculating paste dispensing, according to some embodiments of the invention.

[0012] FIG. 2D is a high-level schematic illustration of a combination of a paste filling unit in a pattern transfer printing (PTP) system, according to some embodiments of the invention.

[0013] FIG. 3A provides an example for multi-layered elements produced by disclosed methods and systems, according to some embodiments of the invention.

[0014] FIGS. 3B and 3C are high-level schematic illustrations of photovoltaic solar cells with multilayered conducting lines, according to some embodiments of the invention.

[0015] FIG. 3D provides examples for dents in a source substrate, filled with paste, according to some embodiments of the invention.

[0016] FIGS. 3E and 3F are high-level schematic illustrations of a receiving substrate with deposited multilayered bump elements, according to some embodiments of the invention.

[0017] FIGS. 4A and 4B are high-level flowchart illustrating pattern transfer methods, according to some embodiments of the invention.

[0018] FIGS. 5A and 5B are high-level schematic side-view illustrations of a pattern transfer process for producing multi-layered features and utilizing a releasing layer, according to some embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0019] In the following description, various aspects of the present invention are described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the present invention. However, it will also be apparent to one skilled in the art that the present invention may be practiced without the specific details presented herein. Furthermore, well-known features may have been omitted or simplified in order not to obscure the present invention. With specific reference to the drawings, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of the present invention only, and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of the invention. In this regard, no attempt is made to show structural details of the invention in more detail than is necessary for a fundamental understanding of the invention, the description taken with the drawings making apparent to those skilled in the art how the several forms of the invention may be embodied in practice.

[0020] Before at least one embodiment of the invention is explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention is applicable to other embodiments that may be practiced or carried out in various ways as well as to combinations of the disclosed embodiments. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting.

[0021] Embodiments of the present invention provide efficient and economical methods and mechanisms for producing multi-layered features using transfer printing processes and thereby provide improvements to the technological field of producing electric circuits. Pattern transfer source substrates and printing methods are described, providing multi-layer paste stack elements like conductor lines or bumps, so that each stack element is printed on a receiving substrate in a single illumination step. The paste is filled layer-by-layer, possibly having different materials in different layers, with layer thickness controlled by parameters of the filling elements, e.g., in case

of blades, the pressure, angle, velocity and flexibility (material) of the blade. Specifically, a bottom layer of the stack may be configured to interface the receiving substrate while one or more top layers may be configured to optimize the quality of the printed features. For example, bottom layers may be configured to bind to the substrate, to modify the substrate (e.g., forming selective emitter (SE) therein) and/or provide a barrier from top layer(s) which may not be compatible with the substrate (e.g., copper on silicon). In case of bumps, different layers may include adhesive and soldering materials. Releasing material may be used to support the single step release of the stack element(s). Advantageously, disclosed pattern transfer sheets and methods enable printing multi-layered conductive lines on PV cells as well as printing multi-layered bumps on PCB, printing resistors and capacitors and other printed electronics devices, and/or providing interposer substrate in advanced packaging of semiconductor chips.

[0022] FIGS. 1, 2A and 2B are high-level schematic illustrations of pattern transfer sheet **100** with a multi-layered stack **150** of printing paste and pattern transfer methods **200**, according to some embodiments of the invention. FIG. 1 illustrates schematically the paste filling and pattern transfer processes, while FIG. 2A illustrates schematically process parameter adjustments for printing multi-layered stack **150** and FIG. 2B illustrates schematically printing multi-layered stack **150** as a stack element with more than two layers. It is noted that for clarity reasons, only two layers **150A**, **150B** are illustrated explicitly; disclosed embodiments however are not limited to two layers, and in various embodiments multi-layered stack **150** may comprise 3, 4, 5 or more layers, prepared and printed according to the principles disclosed herein. While some of the illustrations include for clarity a single trench or dent **110** per sheet **100**, clearly any of the disclosed embodiments is applicable to sheets **100** with multiple trenches or dents **110** arranged in specified patterns on sheet **100**. Furthermore, trenches or dents **110** may comprise trenches, recesses and/or indentations that are embossed (e.g., press molded, pneumatically molded or laser molded) in a similar manner onto sheet **100**, and may have similar or different profiles. For example, trenches or dents **110** may have various profiles (cross section shapes), such as trapezoid, rounded, square, rectangular and/or triangular profiles. In various embodiments, the pattern of trenches **110** on transfer sheet **100** may comprise continuous trenches **110** and/or arrays of separate dents **110**. It is noted that the term “trenches” is not to be construed as limiting the shape of trenches **110** to linear elements, but is understood in a broad sense to include any shape of trenches or dents **110**. One or several trenches or dents **110** may be filled in one or more process step, and material from one or several trenches or dents **110** may be released in the single illumination step, as disclosed herein.

[0023] Pattern transfer sheet **100** comprises a plurality of trenches or dents **110** arranged in a specified pattern (pattern not shown and may be of any configuration). Trenches or dents **110** are configured to be filled with printing paste and consecutively enable and support releasing the printing paste from trenches or dents **110** onto a receiving substrate **70** upon illumination by a laser beam **80**. Pattern transfer sheet **100** comprises a source substrate that may be flexible (e.g., a polymer sheet) or rigid (e.g., a glass substrate). The supporting portion of the pattern transfer sheet **100** may be a stretchable substrate with certain stretching properties such as a polymer substrate, or the supporting portion of the pattern transfer sheet **100** may be a rigid substrate such as a glass substrate. For manufacturing a glass source substrate with respective trenches or dents **110**—several technologies may be applied, for example LIDE® (Laser Induced Deep Etching), which is available from Vitrion, a brand of LPKF Laser & Electronics SE. In non-limiting embodiments, LIDE® technology was used to produce high quality trenches and dents **110** of very small size—from several microns to several tens of microns. It is noted that both polymer and glass source substrates have inherent advantages and disadvantages when used as pattern transfer sheets **100**, and the specific material may be selected according to specific use cases. For example, polymer films are simpler and cheaper in manufacturing while glass substrates are more accurate in pattern dimensions and enable much more cycles of multiple use.

[0024] The printing paste comprises multi-layered stack **150** made of a plurality of layers (**150A**,

150B, and so forth) that is filled **210** by one or more heads **145** to yield filled trenches or dents **115**, and is transferred as stack line(s), bump(s) or other element(s) onto receiving substrate **70** using a single illumination (pattern transfer) step **220**. Filling the printing paste into trenches or dents **110** (stage denoted **210** may be carried out in a consecutive manner, layer-by-layer—as illustrated schematically.

[0025] In the illustrated non-limiting example, two layers **150** of printing paste are filled into trenches or dents **110** consecutively, with a first paste material filled as layer **150A** into trenches or dents **110**, e.g., using a doctor blade filling head **145A** with controlled scooping and removing of paste from the top part of trenches or dents **110**, e.g., carried out by blade **145A** of the same head or alternatively by an additional moving blade (not illustrated). The depth of scooping maybe controlled, e.g., by the scooping blade material, blade angle, head pressure and head movement velocity. A second paste material may be filled on top of layer **150A** as layer **150B**, filling the remaining volume of trenches or dents **110** to yield filled trenches or dents **115** in case of two-layered implementations. Filling layer **150B** with the second paste material using a doctor blade filling head **145B** (followed by minimal or no scooping).

[0026] Consecutive pattern transfer **220** using illumination **80** of pattern transfer sheet **100** from the back side thereof releases stack element(s) **150** of the multi-layered paste from filled trenches or dents **115** onto substrate **70** (e.g., wafer **70**) in a single stage (illustrated in FIG. **1** schematically, with sheet **100** facing down for releasing the paste)—to yield multi-layered elements **160** such as elements in electronics design, lines, fingers or bumps, made of layers **160B**, **160A** of the second and first materials, in opposite order. Second paste layer **150B** becomes bottom layer **160B** contacting receiving substrate **70**, while first paste layer **150A** becomes top layer **160A** of feature **160**. It is noted that while inner layer **150A** typically contacts most of the surface area of trenches or dents **110** in source substrate **100**, bottom layer **160B** typically provides the interface between feature **160** (formed from transferred stack line **150**) and receiving substrate **70**.

[0027] In various embodiments, more than two layers may be filled **240** in to trenches or dents **110** and consecutively transfers, using single stage **250**, onto receiving substrate **70** to yield multi-layered features. For example, three, four, five or more layers may be part of multi-layered stack element(s) **150**—yielding multi-layered elements **160**. In certain embodiments, the thickness of bottom layer **150A** may be between 1-5 μm . Top layer **150B** may be several μm thick, e.g., within any of the ranges 3-5 μm , 4-5 μm , 4-7 μm , 5-10 μm , over 7 μm , or any subranges thereof.

[0028] It is noted that any of heads **145** may comprise a recirculating paste dispensing head as described in Chinese Patent Applications Nos. 202110673006.5 and 202121350578.1, incorporated herein by reference in their entirety. For example, heads **145** which are used to scoop an upper portion of the filled material (see, e.g., in following FIGS. **2A**, **2B**) may comprise doctor blades configured to provide the required scooping, while head **145** filling the last layer (without scooping) may be a recirculating paste dispensing head.

[0029] FIG. **2A** further illustrates schematically various ways to control the thickness of the layers in stack **150** and consequently in feature **160**, according to some embodiments of the invention. For example, layer thickness may be controlled by any of the pressure apply by blades **145**, the angle of blades **145**, the velocity of blades **145** and/or the material blades **145** are made of. Specifically, the thickness of applied layer (denoted schematically **H1**, e.g., layer **150A**) decreases with higher blade pressure, with larger blade angle, with higher blade velocity and with softer blade material—leaving more volume to be filled by consecutive layer(s) **150B** (the volume thickness denoted schematically **H2**). As illustrated schematically in FIG. **1**, the transfer of stack element **150** onto substrate **70** is carried out by illumination applied to the back of sheet **100**, placed directly above and in close proximity to substrate **70**.

[0030] In the illustrated examples, increasing blade pressure (**P**) of blade **145A** filling in layer **150A** from **P**=2.5 bar to **P**=3.5 bar (while keeping the blade pressure of blade **145B** filling in layer **150B** at 2 bar) reduced the thickness of layer **150A**, leaving more volume to form thicker top layer **150B**

(illustrated schematically as filled trenches or dents **115A** and **115B** for lower and higher blade pressure, respectively). Correspondingly, in deposited features layer bottom **160A** became thicker and top layer **160B** became thinner as the filling blade pressure was increased. The reduction in thickness of layer **150A** with increased blade pressure was found to result from enhanced scooping with higher blade pressures, e.g., using Heraeus™ 9651B paste, P=2.5 bar blade pressure of blade **145A** removed about 1 μm of the filled paste (**115A**, leaving 1 μm for layer **150B**) while P=3.5 bar blade pressure of blade **145A** removed about 4 μm of the filled paste (**115B**, leaving 4 μm for layer **150B**). It is noted that the lower pressure of blade **145B** yields a thin and minimally-scooped layer **150B** on top of layer **150A**.

[0031] In the illustrated examples, increasing blade angle (α) of blade **145A** filling in layer **150A** from $\alpha=30^\circ$ to $\alpha=120^\circ$ (under blade pressure of P=2.5 bar, while keeping the blade pressure of blade **145B** filling in layer **150B** at 2 bar and its angle at $\alpha=30^\circ$) reduced the thickness of layer **150A**, leaving more volume to form thicker top layer **150B** (illustrated schematically as filled trenches or dents **115A** and **115B** for lower and higher blade angles, respectively). Correspondingly, in deposited features layer bottom **160B** became thicker and top layer **160A** became thinner as the filling blade angle was increased. The reduction in thickness of layer **150A** with increased blade angle was found to result from enhanced scooping with higher blade angles, e.g., using Heraeus™ 9651B paste, $\alpha=30^\circ$ blade angle of blade **145A** removed about 1 μm of the filled paste (**115A**, leaving 1 μm for layer **150B**) while $\alpha=120^\circ$ blade angle of blade **145A** removed about 4 μm of the filled paste (**115B**, leaving 4 μm for layer **150B**). It is noted that the lower angle of blade **145B** yields a thin and minimally-scooped layer **150B** on top of layer **150A**.

[0032] In the illustrated examples, increasing blade velocity (V) of blade **145A** filling in layer **150A** from V=25 mm/sec to V=100 mm/sec (under blade pressure of P=2.5 bar and blades' angle of $\alpha=30^\circ$ while keeping the blade pressure of blade **145B** filling in layer **150B** at 2 bar) reduced the thickness of layer **150A**, leaving more volume to form thicker top layer **150B** (illustrated schematically as filled trenches or dents **115A** and **115B** for lower and higher blade velocity, respectively). Correspondingly, in deposited features layer bottom **160B** became thicker and top layer **160A** became thinner as the filling blade velocity was increased. The reduction in thickness of layer **150A** with increased blade velocity was found to result from enhanced scooping with higher blade velocities, e.g., using Heraeus™ 9651B paste, V=25 mm/sec blade velocity of blade **145A** removed practically none of the filled paste (**115A**, leaving a minimal volume for layer **150B**) while V=100 mm/sec blade velocity of blade **145A** removed 3-4 μm of the filled paste (**115B**, leaving 3-4 μm for layer **150B**). It is noted that the lower velocity of blade **145B** yields a thin and minimally-scooped layer **150B** on top of layer **150A**. Intermediate values of the blade velocity yield intermediate values of the scooping.

[0033] In the illustrated examples, increasing blade material flexibility (M) of blade **145A** filling in layer **150A** from standard metal (e.g., stainless steel) blades **145A** over plastic (e.g., polyester or ultra-high molecular weight polyethylene—UHMWPE) blades **145A** to rubber blades **145A** (under the same blade pressure, blade angle and blade velocity) reduced the thickness of layer **150A**, leaving more volume to form thicker top layer **150B** (illustrated schematically as filled trenches or dents **115A** and **115B** for medium and high blade flexibility, respectively). Correspondingly, in deposited features layer bottom **160B** became thicker and top layer **160A** became thinner as the filling blade flexibility was increased. The reduction in thickness of layer **150A** with increased blade flexibility was found to result from enhanced scooping with higher blade flexibility, e.g., using Heraeus™ 9651B paste, metal blade **145A** removed practically none of the filled paste (**115A**, leaving a minimal volume for layer **150B**), plastic blade **145A** removed 3-4 μm of the filled paste (**115B**, leaving 3-4 μm for layer **150B**) and rubber blade **145A** removed 6-8 μm of the filled paste (filled trench **115C**, leaving 6-8 μm for layer **150B**). It is noted that the lower flexibility of blade **145B** yields a thin and minimally-scooped layer **150B** on top of layer **150A**.

[0034] It is noted that specific combinations of any of the disclosed parameters and specific

parameter values may be selected with respect to requirements related to the specific pastes used and pre-defined layer structure of stacks **150**. The parameters and their values may be optimized in pre-production steps of the respective process development.

[0035] In contrast to current screen-printing methods, the disclosed methods and systems are used to print ultra-fine metal paste lines on silicon wafers in PV cells, printed electronics, VLSI (very large-scale integration) packaging, etc.—utilizing the advantages of PTP methods of non-contact two process steps printing which separates paste filling stage **140** from pattern transfer stage **250**. Multi-layered fingers as elements **160** enable printing multi-material elements **160**.

Advantageously, in disclosed methods and systems all the layers are formed within the same element (e.g., trench or bump) so they are self-aligned; and as both steps of filling and printing are carried out in the same PTP system with the same cycle time as in case of single layer printing—disclosed printing of multi-layered features becomes cost-effective and enables a very high throughput.

[0036] For example, disclosed methods and systems may be used to print silver stack fingers **160** for PV cells, which enable further reduction of feature width (e.g., to reduce shading losses of the PV cells) while maintaining or even improving conductivity, e.g., (i) not increasing or even decreasing contact resistance (to the silicon substrate) by using bottom layer **160B** made of a paste composition that is optimized for forming the best electrical contact and good adhesion to silicon substrate (e.g., through the silicon nitride layer in TOPCon (tunnel oxide passivated contact or varieties thereof), PERC, (passivated emitter rear contact) etc. PV cells), e.g., by increased concentration of glass frits and/or (ii) not increasing or even decreasing linear resistance (along the finger length) by using top layer **160A** made of a paste composition that is optimized for conductivity along the finger, irrespective of the prior art constraint of having to maintain low surface resistance. Hence, printing elements **160** with one material **160B** interfacing substrate **70** and optimized therefor, and with another material **160A** providing the design's structural and electrical requirements and optimized therefor—enables further narrowing of the printed features while possibly even enhancing their performance such as cell efficiency.

[0037] In another example, disclosed methods and systems may be used to print self-aligned selective emitter (SE) lines, which is presently a major challenge of the PV industry. Specifically, as currently SE lines in silicon substrate are created in a separate process, before the finger metallization, accurate alignment between the printed fingers and the SE lines is required. Current solutions involve printing much wider SE lines (e.g., 100 μm wide) to be covered by much narrower metal fingers (e.g., 30 μm wide)—resulting in additional losses of efficiency in short wavelength spectral range. In contrast, disclosed methods and systems may be used to print SE lines by using bottom layer **160B** that includes the impurities needed for forming the SE lines and using top layer **160A** made of the paste composition that is required for finger metallization—making the present two-step printing process into a one-step printing process and providing the narrowest possible SE lines that exactly fit the metal fingers on top of the SE lines. Consecutively, during the firing process (high temperature heat treatment), all three of the SE lines, the contact to the SE lines and the metal fingers (undergoing silver sintering) are created simultaneously. This application may be applicable, e.g., for PV cells having front metallization over silicon nitride layer, such as TOPCon, PERC and the like. Feasibility of such process was reported in the prior art (e.g., Rohatgi et al. 2001, Self-aligned self-doping selective emitter for screen printed silicon solar cells, presented at the 17th European Photovoltaic Solar Energy Conference and Exhibition; Munich, Germany; Oct. 22-26, 2001), however screen printing does not enable forming multi-layer features in one printing process.

[0038] In yet another example, disclosed methods and systems may be used to print copper metal fingers instead of silver metal fingers, e.g., for PV cells. While copper is basically preferable to silver as metal finger material due to cost and availability considerations, currently using copper requires forming a barrier and/or seed layer (e.g., using a costly sputtering process) below the

copper (e.g., using an electroplating process) to avoid diffusion of copper atoms into silicon substrate **70**—resulting in an expensive and complex production process. In contrast, disclosed methods and systems may be used to print bottom layer **160B** made of a paste composition for providing a barrier layer (e.g., made of any of silver, nickel, titanium, tantalum, etc., combinations and/or alloys thereof) and top layer **160A** made of a copper paste composition. Hence, printing elements **160** with very thin barrier layer **160B** interfacing substrate **70** and relatively thick copper layer **160A** enables using copper fingers in a single printing step—providing significant improvement over the prior art practices. In certain embodiments, layer **160B** may comprise a seed layer, used e.g., for electro-plating of a main conductive layer, e.g., made of copper. In certain embodiments, as illustrated e.g., in FIG. 2B, multilayer stack **150** may comprise barrier layer **150C** (contacting substrate **70** as layer **160C** of deposited line **160**), seed layer **150B** (deposited as layer **160B** on top of layer **160C**) and main conductive layer **150A** on top thereof (deposited as layer **160A** on top of layer **160B**)—forming corresponding element(s) **160** on receiving substrate **70**, such as line(s) and/or bump(s).

[0039] In yet another example, disclosed methods and systems may be used to print bumps composed of two or more layers for applications in the field of advanced packaging of semiconductor chips, e.g., as at least a part of an interposer substrate between the receiving substrate and the chip(s). When printing bumps, the prior art challenge of aligning multiple materials to form the same bump is great, as both dimensions of the bumps are very small. In contrast, the disclosed methods and systems are used to print multi-layered metal paste bumps on various substrates for various uses.

[0040] FIG. 2B schematically illustrates multi-layered stacks **150** with more than two layers—for printing multi-layered features **160** with more than two layers. The thickness of each layer **150A**, **150B**, **150C** (in the non-limiting case of three layers) may be controlled by any of the pressure apply by blades **145**, the angle of blades **145**, the velocity of blades **145** and/or the material blades **145** are made of. Specifically, the thickness of each applied layer (e.g., H1 for layer **150A** and H3 for layer **150B**) decreases with higher blade pressure, with larger blade angle, with higher blade velocity and with softer blade material—leaving more volume to be filled by consecutive layer **150B** over layer **150A** (denoted schematically H2) and by consecutive layer **150C** over layer **150B** (denoted schematically H4). As illustrated schematically in FIGS. 1 and 2A, the transfer of multi-layered stack lines **150** onto substrate **70** is carried out by illumination applied to the back of sheet **100**, placed directly above and in close proximity to substrate **70**.

[0041] FIG. 2C is a high-level schematic illustration of a combination of paste filling heads **145** including doctor blade dispensing **145A** and recirculating paste dispensing **145B**, according to some embodiments of the invention. The example is schematic, and various combinations of one or more doctor blade dispensing heads **145A** and/or recirculating paste dispensing heads **145B** may be used to fill the paste layers **150A**, **150B**, etc. into trenches **110**. For example, recirculating paste dispensing heads **145B** may be used to fill the top paste layer, without scooping.

[0042] Paste dispensing apparatus **170** comprises a printing head **145** comprising at least two feeding openings (at the regions of junctions **177**, **173**), an internal cavity **148** and at least one dispensing opening **146** that are in fluid communication (see, e.g., Chinese Patent Applications Nos. 202110673006.5 and 202121350578.1, incorporated herein by reference in their entirety) and a pressurized paste supply unit **180** configured to circulate paste through printing head **145**. The pressure in the pressurized paste supply unit **180** may be adjusted to maintain continuous circulation of the paste through the feeding openings and internal cavity **148**, and to control dispensing of the paste through dispensing opening(s) **146**. For example, pressurized paste supply unit **180** may comprise a pressurized paste reservoir **174** and a paste pump **172** in fluid communication with internal cavity **148** of printing head **145**, which are configured to circulate the paste therethrough. In non-limiting examples, paste pump **172** may comprise rotating pressure-tight displacement systems with self-sealing, rotor/stator designs for dispensing precise volumes

such as eco-PEN450™ from Dymax™.

[0043] It is noted that printing heads **145** may comprise one or more doctor blade dispensing head(s) **145A** and/or one or more recirculating paste dispensing head(s) **145B**. For example, one, two or more of printing heads **145A**, **145B**, **145C** (see, e.g., FIGS. **1**, **2A**, **2B**) may comprise recirculating paste dispensing head(s) such as paste dispensing apparatus **170** illustrated schematically in FIG. **2C**. Alternatively or complementarily, one, two or more of printing heads **145A**, **145B**, **145C** may comprise doctor blade dispensing head(s) or other types of dispensing heads.

[0044] In various embodiments, paste dispensing apparatus **170** comprises at least one pressure sensor **178** configured to measure the pressure of the circulating paste, at one or more positions in pressurized paste supply unit **180**. Paste dispensing apparatus **170** may further comprise at least one controller (not shown) configured to adjust the pressure in pressurized paste supply unit **180** (or its components) with respect to the measured pressure of the circulating paste. Paste dispensing apparatus **170** may further comprise one or more paste mixer(s) **176** configured to mix the circulating paste. For example, paste mixer(s) **176** may be a static mixer, mixing the paste by utilizing its pressurization. In non-limiting examples, paste mixer(s) **176** may comprise plastic disposable static mixers such as GXF-10-2-ME™ from Stamixco™ made of large diameter plastic housing that includes multiple mixing elements.

[0045] Pressurized paste supply unit **180** may be further configured to introduce the paste into internal cavity **148** via at least one entry opening and to receive the circulated paste via at least one exit opening in printing head **145**. Typically, the entry opening(s) and the exit opening(s) are at the top of printing head **145**, opposite to dispensing opening(s) **146** which faces the substrate onto which the paste is deposited. Alternatively, or complementarily, the entry opening(s) and/or the exit opening(s) may be positioned on sides and/or extension(s) of printing head **145**.

[0046] Pressurized paste supply unit **180** may comprise pressure-controlled paste reservoir **174**, paste pump **172** and mixer **176** that are in fluid communication. Pressure-controlled paste reservoir **174** may be configured to deliver paste to paste pump **172**, which may be configured to deliver the paste through mixer **176** to the entry opening(s). Pressurized paste supply unit **180** may be further configured to mix paste from the exit opening(s) with the paste delivered from pressure-controlled paste reservoir **174** to paste pump **172**. The exit of the pressure-controlled paste reservoir **174** is in fluid communication with the entry of the paste pump **172**. For example, paste in paste reservoir **174** may be delivered to paste pump **172**, mixing with paste exiting from the exit opening(s) of printing head **145**, to be pumped by paste pump **172** into mixer **176**. Paste from mixer **176** may be delivered to the entry opening(s) of printing head **145**, wherein the paste moves along internal cavity **148** and some paste may be dispensed through dispensing opening(s) **146** to fill trenches or dents **110**. The remaining paste may then be mixed with paste from paste reservoir **174** (e.g., delivered through nozzle(s) at junction **173**) to compensate for the dispensed amount, and the paste is circulated through paste dispensing apparatus **170** to maintain its mechanical characteristics and support continued mixing of the paste to maintain its chemical composition. In certain embodiments, paste dispensing apparatus **170** may be further configured to modify paste composition, e.g., by adding additives such as solvents to keep the paste homogenized, possibly in relation to the monitored pressures throughout paste dispensing apparatus **170**. For example, additives such as solvents may be added to the paste entering mixer **176** if needed.

[0047] In various embodiments, pressure-controlled paste reservoir **174** and paste pump **172** may open adjacently to the exit opening(s) of printing head **145** and paste dispensing apparatus **170** may comprise a conduit **175** connecting the exit of mixer **176** to the entry opening(s) of printing head **145**. In some embodiments, pressure-controlled paste reservoir **174** and paste pump **172** may open adjacently to the exit opening(s) of printing head **145**, mixer **176** may be adjacent to the entry opening(s) of printing head **145**, and conduit **175** may connect paste pump **172** to mixer **176**. The dimensions and orientations of paste reservoir **174** and paste pump **172** may vary, e.g., both paste

reservoir **174** and paste pump **172** may be set perpendicularly to printing head **145**, or one or both of paste reservoir **174** and paste pump **172** may be set at an angle to printing head **145**. For example, paste pump **172** may be set obliquely to spread its weight more evenly over printing head **145**. In various embodiments, conduit **175** may be adjusted to conform to any arrangement of paste reservoir **174**, paste pump **172** and mixer **176**, so as to make paste dispensing apparatus **170** more compact or adjust it to a given space and weight distribution requirements within the printing machine.

[0048] In various embodiments, printing head **145**, internal cavity **148** and dispensing slit **146** limited by slit edges **147** (e.g., metallic slit lips **147**) may be elongated and configured with respect to paste properties (e.g., viscosity values), specified throughput and specified features (e.g., length, width and optionally cross section) of the lines or other elements that are to be dispensed by printing head **145**. In certain embodiments, dispensing opening **146** may comprise one or more slits, one or more opening, a plurality of linearly-arranged openings, e.g., one or more lines of circular or elliptical openings, and so forth.

[0049] FIG. 2D is a high-level schematic illustration of a combination of a paste filling unit **102** in a pattern transfer printing (PTP) system **101** (see also FIG. 5B), according to some embodiments of the invention. Paste filling unit **102** is configured to fill patterned trenches or dents **110** on delivered pattern transfer sheets **100** with conductive printing paste, utilizing two or more paste dispensing heads **145** which are applied consecutively to fill trenches or dents **110** with stack **150** made of layers **150A**, **150B**, etc., that is transferred as stack element(s) **150** by PTP system **101** onto receiving substrate **70** using the single illumination step to form multi-layered feature **160** as described herein.

[0050] Two or more paste dispensing heads **145** may comprise respective blades **145A**, **145B**, etc., one for filling each of layers **150A**, **150B**, etc., with at least one of paste dispensing heads **145** being adjustable with respect to the pressure, the blade angle, the blade velocity and/or the blade flexibility as described herein—to adjust the thickness of the respective layers. At least one of paste dispensing heads **145** may be a recirculating paste dispensing head such as paste dispensing apparatus **170** with pressurized paste supply unit **180**, used, e.g., to fill the top layer of stack **150**. Paste filling unit **102** may further comprise a coating unit **103** (illustrated schematically, see Chinese Patent Application No. 202111233805.7, incorporated herein by reference in its entirety—for more details) configured to coat trenches or dents **110** with a releasing layer (see FIGS. 5A, 5B) before filling the multi-layered printing paste into trenches or dents **110**.

[0051] FIG. 3A provides an example for multi-layered elements **160** produced by disclosed methods and systems, according to some embodiments of the invention. The illustrated example in FIG. 3A shows a feature made of 12 μm copper paste **160A** on top of 2 μm silver paste **160B** (the image is focused on the Ag/Cu boundary to emphasize its multi-layered structure). The feature was prepared by first filling into trench **110** copper paste as material **150A** using blade **145A** at $\alpha=30^\circ$ and $P=2.5$ bar and consecutively filling silver paste as material **150B** using blade **145B** at $\alpha=30^\circ$ and $P=2$ bar—and transferring the multi-layered paste from filled trench **115** onto substrate **70** (a silicon wafer) to yield the depicted line.

[0052] FIGS. 3B and 3C are high-level schematic illustrations of photovoltaic solar cells **72** with multilayered conducting lines **160**, according to some embodiments of the invention. In various embodiments, solar cells **72** may comprise multilayered conducting lines **160** produced in a single transfer (illumination) step by pattern transfer method disclosed herein, e.g., using disclosed pattern transfer sheets **100**. For example, one or more multilayered conducting lines **160** may be used in photovoltaic solar cells **72** as conductive line(s) used to deliver electricity produced by photovoltaic solar cells **72** by conversion of light, e.g., with receiving substrate **70** comprising any type of light-converting material such as semiconducting material (e.g., various types of mono- or polycrystalline silicon, various thin film materials, configured as single or multi junction cells) or used in any other photovoltaic technology that utilizes conducting lines.

[0053] FIG. 3D provides examples for dents **110** in source substrate **100**, filled with paste **150**, according to some embodiments of the invention. FIG. 3D thus indicates the feasibility of implementing disclosed systems and methods with dents **110** to produce bumps **160**.

[0054] FIGS. 3E and 3F are high-level schematic illustrations of receiving substrate **70** with multilayered conducting elements **160**, according to some embodiments of the invention. In various embodiments, solar cells **72** may comprise multilayered conducting bumps **160** produced in a single transfer (illumination) step by pattern transfer method disclosed herein, e.g., using disclosed pattern transfer sheets **100**. For example, one or more multilayered conducting elements **160** may be used for providing dense contact matrices for advanced packaging of semiconductor dies, with the multi-layer structure **160** enabling, for example, to combine a material (e.g., in layer **160B** of FIG. 3E or layer **160C** of FIG. 3F) providing enhanced adhesion to substrate **70** (e.g., flux materials that dissolve oxides from the surface of substrate **70** and facilitate wetting thereof) with a material (e.g., in layer **160A** of FIG. 3E or FIG. 3F) that improves soldering ability. Multilayered conducting elements **160** may provide at least a part of an interposer substrate between the receiving substrate and silicon chip(s), e.g., as semiconductor IC packaging connections.

[0055] Multilayered conducting elements **160** may comprise at least two layers **160A**, **160B**, **160C** etc. made of different materials that are transferred as multilayered stack element(s) **150** comprising at least two different types of metal paste **150A**, **150B**, **150C** etc., not respectively, as disclosed herein. Following the single-step transfer, multilayered conducting lines **160** may be processed thermally according to the type(s) of paste used (e.g., fired, cured, sintered, annealed together, etc.) to form the finished lines. As non-limiting examples, silver pastes may be sintered (e.g., dried, heated up to about 800° C. and thereby thermally processed, typically over tens of seconds), silver-epoxy pastes may be cured (e.g., heated up to 200°-300° C. and thereby thermally cross-linked, typically over tens of minutes), copper nano-particle pastes may be sintered at relatively low temperature, and so forth. The specific parameters of the thermal processing of transferred multilayered stacks **B** into multilayered conducting lines **160**, such as temperature profile, duration and the gas atmosphere may be adjusted with respect to the metal paste components in the stack.

[0056] In certain embodiments, multilayered conducting element(s) **160** may comprise bottom layer (e.g., denoted **160B** in FIGS. 3B and 3E, and **160C** in FIGS. 3C and 3F) comprising a material configured to bind (mechanically and/or electrically) transferred stack element(s) **160** to receiving substrate **70**, and at least one top layer (e.g., denoted **160A** in FIGS. 3B and 3E, and **160A**, **160B** in FIGS. 3C and 3F) comprising a material configured to provide conductivity along transferred stack element(s) **160**. For example, the bottom layer may comprise a paste composition that is optimized for forming the best electrical contact and to provide good adhesion to silicon substrate, e.g., having increased concentration of glass frits and/or optionally decreased linear resistance (along the finger length), which is compensated for by the higher linear conductivity of the top layer.

[0057] In certain embodiments, multilayered conducting line(s) **160** may comprise bottom layer (e.g., denoted **160B** in FIG. 3B, **160C** in FIG. 3C) comprising a material configured to form selective emitter (SE) lines **160** on receiving substrate **70**, and at least one top layer (e.g., denoted **160A** in FIG. 3B, **160A**, **160B** in FIG. 3C) comprising a material configured to provide conductivity along transferred stack line **160** and electrical contact to receiving substrate **70**. For example, the bottom layer may comprise a paste composition that provides the impurities needed for forming the SE lines and compensating for possible reduced conductivity thereof by the higher linear conductivity of the top layer—allowing the production of much thinner SE lines as they are transferred using a single PTP step.

[0058] In certain embodiments, multilayered conducting element(s) **160** may comprise bottom layer (e.g., denoted **160B** in FIGS. 3B and 3E, and **160C** and optionally **160B** in FIGS. 3C and 3F) comprising a material configured to provide a barrier and/or seed layer upon receiving substrate **70**,

and at least one top layer (e.g., denoted **160A** in FIGS. 3B and 3E, and **160A** in FIGS. 3C and 3F) comprising a material configured to provide conductivity along or to transferred stack element(s) **160**. For example, the bottom layer may comprise paste of, e.g., silver, nickel, titanium, tantalum, etc., combinations and/or alloys thereof) as a barrier/seed layer, with the top layer comprising a copper paste that provides high conductivity at reduced material costs. Barrier/seed layer(s) may comprise one, two or more layers, as illustrated e.g., schematically in FIGS. 3C and 3F (with top layer **160A** comprising copper).

[0059] In various embodiments, photovoltaic solar cells **72** may comprise combinations of multilayered conducting lines and/or other multilayered conducting elements **160** disclosed herein.

[0060] FIGS. 4A and 4B are high-level flowchart illustrating pattern transfer methods **200**, according to some embodiments of the invention. The method stages may be carried out using pattern transfer sheet(s) **100** and/or with respect to corresponding PTP systems, which may optionally be configured to implement method **200**.

[0061] Method **200** comprises filling with printing paste a plurality of trenches or dents that are arranged in a pattern on a source substrate (stage **210**), wherein the printing paste comprises a stacked plurality of layers and the filling is carried out layer-by-layer, and releasing the printing paste from the trenches or dents onto a receiving substrate through illumination by a laser beam—transferring the whole stack elements (e.g., line or bump) of multi-layered printing paste onto the receiving substrate using a single illumination step (stage **220**).

[0062] Filling **210** may be carried out by filling the trenches or dents with different pastes of different thickness until full filling of the trenches or dents in the source substrate (stage **212**). For example, filling **210** may be carried out by a plurality of blades, one for each layer, and the pattern transfer method further comprising adjusting a thickness of at least one of the layers by adjusting at least one of: a pressure applied to the respective blade, a blade angle of the respective blade, a blade velocity of the respective blade, and/or a flexibility of the respective blade. For example, the pressure may be selected within 2-4 bar, the blade angle may be selected within 30°-120°, the blade velocity may be selected within 25-100 mm/sec and/or a blade material is selected from stainless steel, plastic and rubber.

[0063] In some embodiments, at least two of the layers comprise different material compositions and/or different materials, e.g., selected from: (i) a bottom layer comprising a material configured to bind, mechanically and/or electrically, the transferred stack element(s) to the receiving substrate and at least one top layer comprising a material configured to provide conductivity along the transferred stack element(s), (ii) a bottom layer comprising a material configured to form selective emitter (SE) lines on the receiving substrate, and at least one top layer to provide conductivity along the transferred stack line and electrical contact to the receiving substrate, and/or (iii) a bottom layer comprising a material configured to provide a barrier and/or seed layer upon the receiving substrate and at least one top layer comprising a material configured to provide conductivity along or to the transferred stack element(s). In certain embodiments, the uppermost layer may comprise a releasing material, e.g., as disclosed below.

[0064] Method **200** may further optionally comprise applying a releasing layer to facilitate full stack printing (stage **225**), illustrated in more detail in FIG. 4B and comprising, for example, (i) prior to the filling, coating the trenches or dents internally with a releasing layer material and drying the releasing layer material to form a solid coating of the trenches or dents configured to disintegrate upon illumination—to facilitate release of the stack element(s) of paste layers filled within the coated trenches or dents (stage **230**), optionally, cleaning a surface of the pattern transfer sheet between the trenches or dents from coating residues (stage **232**); and (ii) after printing, cleaning the receiving substrate by removing disintegration products of the coating therefrom (stage **240**).

[0065] Advantageously, disclosed methods and systems, based on non-contact paste transfer, enable printing multi-layered elements, in contrast e.g., to prior art methods such as screen printing

(SP) which is a cost-effective production method for printing single layers of high viscosity pastes on different substrates by paste filling of the screen and printing features on a receiving substrate carried out simultaneously. However, as screen printing is a contact technology, printing multi-layered features requires to transfer a substrate with the first (still wet) layer to a dryer and then to the same or another printer with an additional alignment of the second layer to the previous layer. The accuracy of such alignment is limited (mainly because two different screens are not identical and deviate differently with time), the process is much more costly and it is hardly possible to print very fine multi-layer features using prior art SP.

[0066] FIGS. 5A and 5B are high-level schematic side-view illustrations of a pattern transfer process for producing multi-layered features and utilizing a releasing layer **120**, according to some embodiments of the invention. Trenches **110** in sheet **100** (as source substrate) may be coated internally by coating **120** configured to disintegrate upon illumination **80**, to enhance the releasing of multi-layered paste stack **150**. Coated trenches that are filled with multi-layered paste are denoted schematically by the numeral **115**. During the releasing of multi-layered paste stack element(s) **150**, coating **120** may disintegrate (indicated schematically by numeral **120A**) and be deposited on wafer **70** (as receiving substrate) together with released multi-layered paste stack element(s) **150**. As a result, wafer **70** may comprise the released multi-layered paste stack element(s) (denoted schematically by numeral **150A**) and residues (and/or disintegration products) of coating (denoted schematically by numeral **120B**)—an intermediate state denoted by numeral **130**. Following the pattern transfer process (by PTP system **101**), coating residues **120B** may be removed from wafer **70**, e.g., by high temperature treatment (e.g., sintering or drying), which is usually applied for sintering printed metal paste in PV cells manufacturing, or by using an illumination **82** to modify and/or remove the residues—to clean the receiving substrate and yield wafer **70** with released paste and without coating disintegration products and residues. Illumination **82** may be same as, similar to or different from illumination **80**. In various embodiments, any of the following methods may be used to remove residues **120B** of the releasing material: high temperature decomposition (e.g., firing), selective laser vaporization, vacuum-assisted vaporization, air blowing and/or solvent wet washing.

[0067] Coating **120**, also termed releasing layer **120** for its function, may be a thin layer (e.g., be 1-10 μm thick) deposited on pattern transfer sheet **100** as a source substrate or at least into trenches or dent(s) **110** thereof. In some cases, the material(s) of releasing layer **120** (denoted—the releasing material(s)) may be mixed with any of the components of the multilayered paste prior to filling the layers into trenches or dent(s) **110**. Non-limiting examples for releasing materials include acetone-based ClearWeld™ LD920 series and acrylic resins such as Spectra 390' by Epolin.

[0068] For example, coating **120** may be carried out by spreading a coating solution (for example having a solid content between 10% and 20%, or 12%, 14%, 16%, 18%) (e.g., using blade(s), rolls, dies, etc.) and drying the releasing layer to form a solid layer in the trenches or dent(s), e.g., which may range in thickness between 1 μm and 10 μm (for example 3 μm , 5 μm , 7 μm , 9 μm). If required, the surface of pattern transfer sheet **100** between trenches or dent(s) **110** from coating residues, e.g., using scraping blade(s) and/or sticky rollers.

[0069] Coating **120** may comprise organic materials such as NIR absorbing dyes. At least one of the materials for coating **120** is selected to absorb laser illumination **80** used in the PTP process (e.g., laser with a wavelength of 1064 nm, 1070 nm or any other wavelength in the NIR spectral range). One or more of the materials may be selected to have a maximal absorption at the illumination wavelength. The releasing material(s) may be configured to change phase, evaporate and/or be ablated upon absorption of the laser illumination energy—to yield a thrusting force that pushes multi-layered paste stack element(s) **150** out of trenches or dent(s) **110** (without changing the shape of the paste) and releases multi-layered paste stack element(s) **150** onto receiving substrate **70**. For example, for using Nd:Yag laser (1060-1085 nm) for illumination **80**, coating **120** may comprise NIR (near infrared) absorbing dyes.

[0070] Coating **120** may be used to replace or augment the use of volatile components of the paste (e.g., added to evaporate upon illumination and release paste multi-layered paste stack **150**).

[0071] Advantageously, the inventors have found out that using coating **120** enables to print paste pattern on receiving substrate **70** with a much higher aspect ratio, e.g., at least 0.7 and up to 1-2—instead of prior art 0.4-0.5; as well as to print extremely narrow fingers, down to 10 μm wide—instead of prior art above 25-30 μm wide—as illustrated in Chinese Patent Application No. 202111233805.7 filed on Oct. 22, 2021, which is incorporated herein by reference in its entirety. Furthermore, avoiding use of volatile compounds in components of the paste broadens the range of printable pastes and makes the process much less sensitive to drying of the paste within the PTP system. Furthermore, although releasing layer **120** is much thinner than paste, a lower laser power is required to release the paste and therefore printing can be carried out at a higher scanning beam velocity, yielding higher system throughput. Finally, using releasing layer **120** makes the printing process more accurate, avoiding printing quality defects of the printing pattern such as paste debris.

[0072] In various embodiments, as disclosed, e.g., in Chinese Patent Application No.

202111233805.7, incorporated herein by reference in its entirety, coating **120**, functioning as the releasing layer, may be deposited or applied by any known technique, such as gravure, micro gravure, transfer roll, slot extrusion, reverse comma, Mayer rod, doctor blade or other techniques, applied by corresponding coaters. For example, using doctor blade(s), trenches **110** may be filled with a coating solution while keeping the surfaces between trenches **110** clean. After drying the coating solution, coating **120** is formed within trenches **110** at a thickness determined mainly by the solid content of the coating solution. Enclosed chamber doctor blade (protecting or enclosing the coating solution) may be configured to improve the control of the coating solution and to ensure the cleanliness of the surfaces between the trenches. In another example, the coating solution may be applied to the surface of at least a portion of pattern transfer sheet **100**, e.g., using a rod spreading technique, spraying, slot extrusion, etc., and drying to form coating **120** as well as the residues on the surfaces between trenches **110**. The residues may be removed by various methods, such as scraping blade(s) (made, e.g., of metal, plastic, rubber, etc.) set at specified angles, sticky rollers, etc.—leaving coating **120** within trenches **110**. Removed residues may be recycled. Any of the coating processes may be carried out in stationary mode (with stationary pattern transfer sheet **100** and respective coating elements moving with respect thereto) and/or in continuous mode (with stationary coating elements and pattern transfer sheet **100** moving with respect thereto, e.g., using rolls in a roll-to-roll approach). The coating processes may be carried out in advance, before feeding pattern transfer sheet **100** into PTP system **101**, or during their movement in PTP system **101**, e.g., just before paste filling of the trenches.

[0073] In various embodiments, at least the surfaces of trenches or dent(s) **110** may be treated to enhance or control adhesion of coating **120** thereto, using surface treatment techniques such as plasma (e.g., corona), applying silane additives, etc. Surface treatment may be configured to balance between a required coating adhesion within trenches or dent(s) **110**, removal of residues from the surfaces between trenches or dent(s) (if needed) and paste release by coating disintegration, as well as by possible sheet recycling considerations.

[0074] In various embodiments, PTP system **101** may be configured to inspect and control any of the multilayer deposition stages and/or the coating and cleaning processes, e.g., at a print quality station controlling the printing quality of the fingers, to keep the processes within predefined process windows.

[0075] In various embodiments, coating residues **120B** may be removed by any of the following procedures: high temperature decomposition, selective laser vaporization, vacuum assisted vaporization (reducing the boiling point of coating residues **120B**), air blowing and/or solvent washing. Using heat-based procedures to remove coating residues **120B**, the coating material may be selected to have decomposition and/or boiling temperatures that do not affect deposited element(s) such as finger(s) or bump(s) **160** and receiving substrate **70**. For example, coating

material may be selected to have decomposition and/or boiling temperatures that are well below the high sintering temperature of the c-Si PV metallization process (-850°C .), or optionally below the curing temperature for cured pastes (e.g., silver epoxy pastes). For example, the decomposition and/or evaporation temperatures of coating residues **120B** may be in the range of 200°C .- 300°C .. In certain embodiments, same laser illumination **80** used to evaporate layer **120** used to release multi-layered paste stack **150** may also be used to evaporate and/or decompose coating residues **120B** on wafer **70**. It is noted that removing coating residues **120B** may be carried out in a single process stage or in two or more process stages, e.g., high temperature decomposition (e.g., by laser illumination **80** and/or in a drying oven or sintering furnace) followed by low temperature washing or blowing.

[0076] The following non-limiting examples disclose NIR absorbing dyes for use with Nd:Yag laser illumination. In addition to the NIR absorbing dyes, coating **120** may further comprise solvents, as well as optionally binders, surface agents and/or viscosity modifiers.

[0077] Non-limiting examples for components of NIR absorbing dyes include Diimonium ionic complex(es), Ditholene complex(es) and/or phthalocyanine. The illustrated Diimonium ionic complex may include, e.g., alkyl chain(s) as one or more residue(s) R and counter ions (not shown) such as 2SbF_6 .

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[0078] Additional, non-limiting examples for components of NIR absorbing dyes in any of the disclosed embodiments may include any of cyanine (tetramethylindo(di)-carbocyanines) dyes such as, e.g., open chain cyanines ($\text{R}-\text{N}=\text{CH}[\text{CH}=\text{CH}]_n-\text{NR}$), hemicyanines ($\text{Aryl}=\text{N}=\text{CH}[\text{CH}=\text{CH}]_n-\text{NR}$), closed chain cyanines ($\text{Aryl}=\text{N}=\text{CH}[\text{CH}=\text{CH}]_n-\text{N}=\text{Aryl}$), neutrocyanines ($\text{R}_2\text{N}=\text{CH}[\text{CH}=\text{CH}]_n-\text{CN}$ and $\text{R}-\text{N}=\text{CH}[\text{CH}=\text{CH}]_n-\text{CHO}$), e.g., with extended $[\text{CH}=\text{CH}]_n$ chains, or variants or mixtures thereof; phthalocyanine or naphthalocyanine dyes (comprising four isoindole units linked by a ring of nitrogen atoms) or metal complexes thereof (e.g., with aluminum or zinc), dithiolene metal complexes (having one to three dithiolene ligands), e.g., with nickel, squaraine dyes such as squarylium dye III, quinone analogues, diimonium compounds and azo derivatives, and/or any of their variants, derivatives and/or combinations.

[0079] Solvent(s) included in the coating formulation may be selected to dissolve the dye(s) as well as optional binder(s) and additive(s), and may comprise, e.g., acetone, ketones, alcohols, aromatic hydrocarbons and/or glycol ethers solvents.

[0080] The coating formulation may further comprise binder(s) comprising polymers and/or polymer precursors for forming coating **120** as a continuous film in trenches or dent(s) **110**, such as, e.g., poly-vinyl butyral, ethyl-cellulose and/or derivatives thereof, as long as the decomposition temperature of binder(s) is below the paste sintering temperature (e.g., 800°C .. for silver pastes, or e.g., 850°C ., 800°C ., 750°C ., 700°C ., 650°C ., 600°C ., 550°C ., 500°C ., 450°C ., 400°C ., 350°C .. for silver pastes or other metal pastes or mixtures thereof) or below the paste curing temperature (e.g., 300°C .. for silver epoxy pastes, or e.g., 300°C ., 250°C ., 200°C .. for silver epoxy pastes or other metal pastes). Binder(s) may further be selected to minimize the amount of coating residues **120B** and/or to simplify the removal thereof.

[0081] The coating formulation may further comprise surface wetting additive(s) selected to stabilize the dispersion of the different coating components and/or viscosity modifier(s) selected to support the application of coating **120** in trenches or dent(s) **110**. The additive(s) and/or viscosity modifier(s) are selected to have their decomposition temperature below the paste sintering or curing temperature (depending on the type of the paste) and may further be selected to minimize the amount of coating residues **120B** and/or to simplify the removal thereof.

[0082] In certain embodiments, the solid content of coating **120** may be between 10 wt % and 20 wt % to enable reaching coating **120** which is 1-10 μm thick within trenches or dent(s) **110** that are 20-30 μm deep.

[0083] In certain embodiments, the coating formulation or any of its components, such as the NIR absorbing dyes, may be mixed at least into layer **150A**, which as the inner (bottom) layer and commonly the thickest layer, has the largest contact area with trenches or dent(s) **110** of sheet **100**. Mixing releasing material into at least inner layer **150A** of multi-layered paste stack **150** may enhance releasing or possible at least partially enable replacing coating **120** (e.g., in certain regions of pattern transfer sheet **100**). For example, at least a portion of the release material may be mixed with at least inner layer **150A** of paste stack **150** (e.g., as a component of the paste that is deposited as layer **150A** or as an additive that is mixed during past deposition) to improve the releasing of multi-layered paste stack element(s), e.g., line or bump **150**. Mixing releasing material into at least the inner (bottom) layer of paste stack **150** may yield print quality improvement. Disclosed releasing material may be added to provide a few percent of the material of the inner layer of multi-layered paste stack **150**, e.g., between 1-5% of the inner layer material. In certain embodiments, both coating **120** and the mixing of the releasing material into at least the inner layer paste may be applied.

[0084] Accordingly, at least inner layer **150A** of multi-layered paste stack **150** may comprise, in addition to commercially available printing pastes (e.g., SOL9651B™ from Heraeus™), NIR absorbing dyes such as Diimonium ionic complex(es), Ditholene complex(es) and/or phthalocyanine as illustrated herein. It is noted that NIR absorbing dyes may be added to silver pastes, to silver epoxy pastes, or to any other material, which is used as the inner layer of the multi-layer stack within the trench.

[0085] Certain embodiments comprise a multi-layered paste stack for use in a pattern transfer process (that includes filling the paste into trenches in the pattern transfer sheet which are arranged in a specified pattern, and consecutively releasing the paste from the trenches or dent(s) onto the receiving substrate upon illumination by the laser beam). For example, in case the illumination is in NIR, the paste comprises a releasing material configured to enhance the releasing of the paste from the trenches or dent(s), with the releasing material comprising at least one NIR absorbing dye comprising at least one of: a Diimonium ionic complex, a Ditholene complex, phthalocyanine, derivatives, salts and/or combinations thereof. Non-limiting examples include dyes from TCI (Tokyo Chemical Industry, Ltd), Epolight™ 1117 (Tetrakis(decyl)ammonium structure) by Epolin and/or Lunir5™ by Luminochem.

[0086] Disclosed sheets **100** may be used to print fine lines or bumps of thick metallic paste on silicon wafers, e.g., for photovoltaic (PV) cells as well as to produce electronic circuits by creating conductive lines, bumps, pads or other features (e.g., on laminates for PCBs) for printed passive electronic components like resistors or capacitors, or for other printed electronic devices. Other applications may comprise creating conductive features in the manufacturing processes of mobile phones antennas, decorative and functional automotive glasses, semiconductor integrated circuits (IC), semiconductor IC packaging connections, printed circuit boards (PCB), PCB components assembly, optical biological, chemical and environment sensors and detectors, radio frequency identification (RFID) antennas, organic light-emitting diode (OLED) displays (passive or active matrix), OLED illuminations sheets, printed batteries and other applications. For example, in non-limiting solar applications, the metallic paste may comprise metal powder(s), optional glass frits and modifier(s), volatile solvent(s) and non-volatile polymer(s) and/or or resin(s). A non-limiting example for the paste includes SOL9651B™ from Heraeus™. The metal pastes may include silver, copper or other metals and/or their combinations and/or alloys. Any of the metal pastes may include any type of electrically conductive adhesive (ECA) pastes.

[0087] Filling of the paste into trenches or dent(s) **110** may be carried out by any type of paste-filling head operating within any type of PTP system **101**. The filling process may be controlled to ensure continuous and uniform filling of the trenches and marks with paste.

[0088] In certain embodiments, pattern transfer sheet **100** may be transparent to laser beam **80** and comprise at least a top polymer layer **114** comprising trenches or dent(s) **110** (illustrated in FIGS. **1**

and 5A) embossed, press molded, pneumatically molded or laser molded thereon. In the illustrated non-limiting example, trenches or dent(s) **110** are illustrated as being trapezoid in cross section. [0089] In certain embodiments, pattern transfer sheet **100** may be transparent to laser beam **80** and comprise a glass source substrate comprising etched trenches or dent(s) **110** (illustrated in FIGS. **1** and **5A**). In the illustrated non-limiting example, trenches or dent(s) **110** are illustrated as being trapezoid in cross section.

[0090] Periodical trenches or dent(s) **110** may comprise trenches, recesses and/or indentations that are embossed (e.g., press molded, pneumatically molded or laser molded) in a similar manner into top polymer layer **114**, or etched into rigid glass source substrate as pattern transfer sheet **100**, and may have similar or different profiles. For example, trenches or dent(s) **110** may have various profiles (cross section shapes), such as trapezoid, rounded, square, rectangular and/or triangular profiles (for lines) or circular, elliptical, square, rectangular, hexagonal and others (for indentations, or dents). In various embodiments, the pattern of trenches **110** on transfer sheet **100** may comprise continuous trenches **110** and/or arrays of separated dents. It is noted that the term “trenches” is not to be construed as limiting the shape of trenches **110** to linear elements, but is understood in a broad sense to include any shape of trenches **110**. Specifically, cavities **110** that are illustrated schematically in FIGS. **1**, **2A-2C**, **5A** and **5B** as well as their various filling methods illustrated schematically in FIGS. **2A-2D** may be configured as dents **110**, which may be non-linear. For example, dents **110** in source substrate **100** used for printing multi-layer bumps, may be of different shapes like circular, elliptical, square, rectangular, hexagonal and others in order to maximally fit the shape and size of pads on receiving substrate **70** on which bumps **110** are deposited. Non-limiting examples of circular and hexagonal shapes of multi-layered elements **160** produced using corresponding dents having similar shapes are illustrated schematically in FIGS. **3E** and **3F**, respectively.

[0091] Pattern transfer sheet **100** may further comprise a bottom polymer layer **112** having a melting temperature that is higher than an embossing temperature of top polymer layer **114**. In a non-limiting example, top polymer layer **114** may have a melting temperature ($T_{sub.m}$) below 170° C., below 150° C., below 130° C., below 110° C. (or any intermediate range) in case it is made of semi-crystalline polymer, or a glass temperature ($T_{sub.g}$) below 160° C., below 140° C., below 120° C., below 100° C. (or any intermediate range) in case it is made of amorphous polymer. The melting temperature of bottom polymer layer **112** may be above the melting point of top polymer layer **114**, e.g., above 100° C. (e.g., in case top polymer layer **114** is made of polycaprolactone and has $T_{sub.m}/T_{sub.g}$ about 70° C.), above 120° C., above 150° C., above 160° C. (e.g., bi-axially-oriented polypropylene) and up to 400° C. (e.g., certain polyimides), or intermediate values.

[0092] In various embodiments, polymer layers **112**, **114** may be made of at least one of: polyethylene, polypropylene, polyethylene terephthalate, polybutylene terephthalate, polyethylene naphthalate, fully aromatic polyester, other copolymer polyester, polymethyl methacrylate, other copolymer acrylate, polycarbonate, polyamide, polysulfone, polyether sulfone, polyether ketone, polyamideimide, polyether imide, aromatic polyimide, alicyclic polyimide, fluorinated polyimide, cellulose acetate, cellulose nitrate, aromatic polyamide, polyvinyl chloride, polyphenol, polyarylate, polyphenylene sulfide, polyphenylene oxide, polystyrene, or combinations thereof—as long as top polymer layer **114** has a melting or glass transition temperature ($T_{sub.m}/T_{sub.g}$) below the melting or glass transition temperature ($T_{sub.m}/T_{sub.g}$) of bottom polymer layer **112** and/or as long as bottom polymer layer **112** is not affected by the processing conditions of top polymer layer **114**.

[0093] In certain embodiments, the bottom polymer layer **112** and the top polymer layer **114** may be each between 10 μm and 100 μm thick, e.g., between 15 μm and 80 μm thick, between 20 μm and 60 μm thick, between 25 μm and 40 μm thick, or within any intermediate range, with bottom polymer layer **112** being at least as thick as top polymer layer **114**. The bottom polymer layer **112**

and the top polymer layer **114** may be attached by an adhesive layer **113** thinner than 10 μm (e.g., thinner than 8 μm , thinner than 6 μm , thinner than 4 μm , thinner than 2 μm or having any intermediate thickness), that is transparent to laser beam **80**. For example, in certain embodiments, top polymer layer **114** may be thicker than the depth of trenches **110** by several μm , e.g., by 5 μm , by 3-7 μm , by 1-9 μm , or by up to 10 μm . For example, trenches **110** may be 20 μm deep, top polymer layer **114** may be between 20-30 μm thick (e.g., 25 μm thick) and bottom polymer layer **112** may range in thickness between 30 μm and 40 μm (it is noted that thicker bottom polymer layers **112** provide better mechanical performances).

[0094] The temperature and thickness of top polymer layer **114** and bottom polymer layer **112** may be designed so that top polymer layer **114** has good molding, ductility and certain mechanical strength, while bottom polymer layer **112** has good mechanical strength. Both top polymer layer **114** and bottom polymer layer **112** may be designed to have good bonding properties.

[0095] Elements from FIGS. 1-5B may be combined in any operable combination, and the illustration of certain elements in certain figures and not in others merely serves an explanatory purpose and is non-limiting. It is noted that disclosed values may be modified by at least $\pm 10\%$ of the respective values.

[0096] Advantageously, while the prior art cannot achieve a printed pattern with features comprising multiple conductive layers, disclosed embodiments meet the performance and production challenges of paste fingers for electronic circuits using a single-stage multi-layered stack printing, which enables to combine different materials within the stack, separating the substrate compatibility requirement and the finger performance requirement from each other, and inherently ensuring alignment between the layers.

[0097] In the above description, an embodiment is an example or implementation of the invention. The various appearances of “one embodiment”, “an embodiment”, “certain embodiments” or “some embodiments” do not necessarily all refer to the same embodiments. Although various features of the invention may be described in the context of a single embodiment, the features may also be provided separately or in any suitable combination. Conversely, although the invention may be described herein in the context of separate embodiments for clarity, the invention may also be implemented in a single embodiment. Certain embodiments of the invention may include features from different embodiments disclosed above, and certain embodiments may incorporate elements from other embodiments disclosed above. The disclosure of elements of the invention in the context of a specific embodiment is not to be taken as limiting their use in the specific embodiment alone. Furthermore, it is to be understood that the invention can be carried out or practiced in various ways and that the invention can be implemented in certain embodiments other than the ones outlined in the description above.

[0098] The invention is not limited to those diagrams or to the corresponding descriptions. For example, flow need not move through each illustrated box or state, or in exactly the same order as illustrated and described. Meanings of technical and scientific terms used herein are to be commonly understood as by one of ordinary skill in the art to which the invention belongs, unless otherwise defined. While the invention has been described with respect to a limited number of embodiments, these should not be construed as limitations on the scope of the invention, but rather as exemplifications of some of the preferred embodiments. Other possible variations, modifications, and applications are also within the scope of the invention. Accordingly, the scope of the invention should not be limited by what has thus far been described, but by the appended claims and their legal equivalents.

Claims

1. A pattern transfer method comprising: filling, layer-by-layer, a plurality of stack elements that are each made of a plurality of printing paste layers into a plurality of dents on a source substrate,

wherein the stack elements are filled into corresponding dents that are arranged in a pattern, and releasing the plurality of stack elements from the corresponding plurality of dents onto a receiving substrate simultaneously, in a single illumination step by a laser beam, forming transferred stack bumps on the receiving substrate, wherein at least two of the paste layers comprise different material compositions.

2. The pattern transfer method of claim 1, wherein the source substrate comprises a flexible polymer sheet.

3. The pattern transfer method of claim 1, wherein the source substrate comprises a rigid glass substrate.

4. The pattern transfer method of claim 1, wherein the filling is carried out by a plurality of blades, one respective blade for each paste layer, and the pattern transfer method further comprising adjusting a thickness of at least one of the paste layers by adjusting at least one of: a pressure applied to the respective blade, a blade angle of the respective blade, a blade velocity of the respective blade, and/or a flexibility of the respective blade.

5. The pattern transfer method of claim 4, wherein the pressure is selected within 2-4 bar, the blade angle is selected within 30°-120°, the blade velocity is selected within 25-100 mm/sec and/or a blade material is selected from stainless steel, plastic and rubber.

6. The pattern transfer method of claim 1, wherein the at least two of the paste layers on the receiving substrate that comprise the different material compositions yield a configuration on the receiving substrate that comprises a bottom paste layer comprising a material configured to bind the transferred stack elements to the receiving substrate and at least one top paste layer comprising a material configured to provide conductivity along the transferred stack elements.

7. The pattern transfer method of claim 6, wherein the bottom paste layer comprises a releasing material composition.

8. The pattern transfer method of claim 6, wherein an uppermost paste layer comprises a releasing material.

9. The pattern transfer method of claim 1, wherein the at least two of the paste layers on the receiving substrate that comprise the different material compositions yield a configuration on the receiving substrate that comprises a bottom paste layer comprising a material configured to provide a barrier and/or seed layer upon the receiving substrate and at least one top paste layer comprising a material configured to provide conductivity along the transferred stack elements.

10. The pattern transfer method of claim 9, wherein an uppermost paste layer comprises a releasing material.

11. The pattern transfer method of claim 1, further comprising prior to filling the plurality of stack elements, coating the bumps internally with a coating configured to disintegrate upon illumination by the laser beam, to enhance the releasing of the stack elements.

12. The pattern transfer method of claim 11, further comprising, after the releasing, cleaning the receiving substrate by removing disintegration products of the coating therefrom.

13. The pattern transfer method of claim 11, wherein the coating is carried out by spreading a coating solution having a solid content between 10% and 20% and drying thereof.

14. The pattern transfer method of claim 11, wherein the coating is carried out by at least one application technique of: gravure, micro gravure, transfer roll, slot extrusion, reverse comma, Mayer rod and/or doctor blade.

15. The pattern transfer method of claim 11, further comprising the cleaning of the receiving substrate, carried out by at least one of: high temperature decomposition, illumination by the illumination source used to release the stack elements, selective laser vaporization, vacuum assisted vaporization, air blowing and/or solvent washing.

16. The receiving substrate produced by the method of claim 1, having the transferred plurality of stack bumps as released from the corresponding plurality of dents each stack bump being released in the single illumination step by the laser beam.

- 17.** An interposer substrate comprising the receiving substrate with transferred bumps of claim 16.
- 18.** A pattern transfer method comprising: filling, layer-by-layer, a plurality of stack lines that are each made of a plurality of printing paste layers into a plurality of trenches on a source substrate, wherein the stack lines are filled into corresponding trenches that are arranged in a pattern, and releasing the plurality of stack lines from the corresponding plurality of trenches onto a receiving substrate simultaneously, in a single illumination step by a laser beam, forming transferred stack lines on the receiving substrate, wherein at least two of the paste layers comprise different material compositions.
- 19.** The pattern transfer method of claim 18, wherein the source substrate comprises a flexible polymer sheet.
- 20.** The pattern transfer method of claim 18, wherein the source substrate comprises a rigid glass substrate.
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